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CSA1509-Cloud computing and Big Data Analytics

EXPERIMENT – 22

Running Virtual Machines with Different Configurations

Aim:

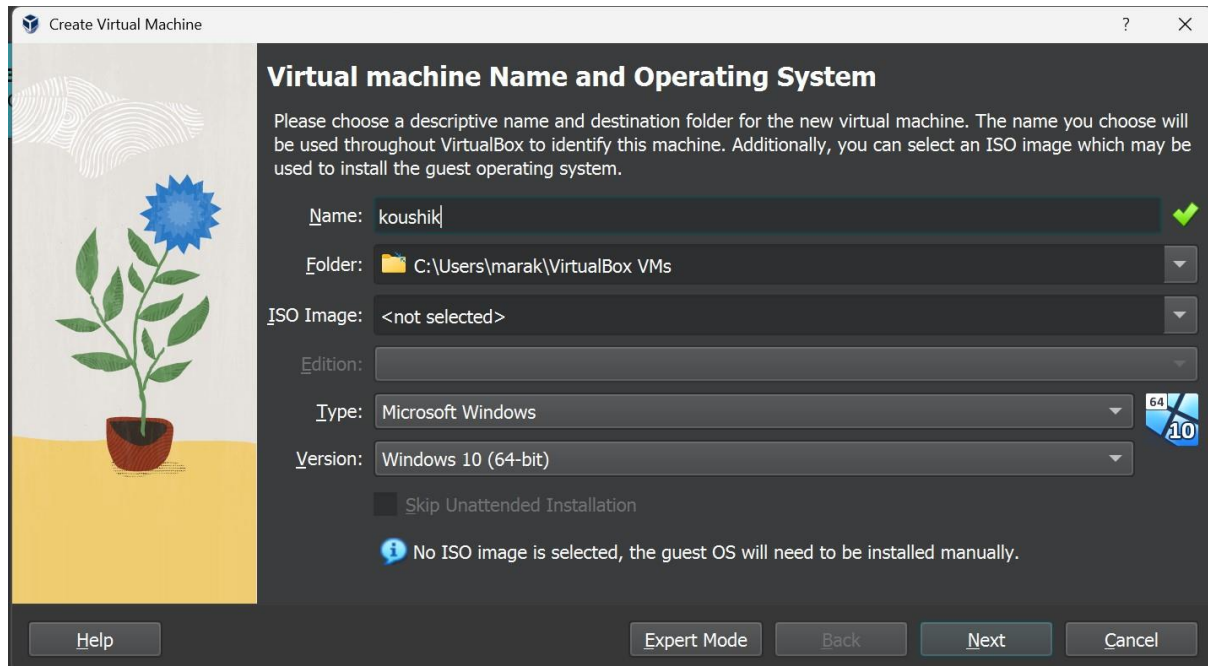
To run virtual machines with different configurations and determine how many virtual machines can be utilized simultaneously based on the available system resources.

Short Procedure:

1. Open Oracle VirtualBox and create a new virtual machine.
2. Configure the VM with different RAM, CPU, and storage settings.
3. Install the required operating system in each virtual machine.
4. Start the virtual machines one by one and observe the system performance.
5. Continue running VMs until the available RAM/CPU resources are nearly utilized.
6. Record the number of VMs running simultaneously and their configurations.
7. Compare the configurations and determine the maximum number of VMs that can be utilized at a particular time. **Result:**

The number of virtual machines that can run simultaneously depends on the host system's available RAM, CPU, and storage resources.

Step 1 : CREATING VM AND PROVIDING ISO FILE



The image shows the 'Create Virtual Machine' window in Oracle VM VirtualBox. The window has a title bar with a question mark and a close button. On the left is a decorative panel with a stylized plant. The main area is titled 'Virtual machine Name and Operating System' and contains instructions: 'Please choose a descriptive name and destination folder for the new virtual machine. The name you choose will be used throughout VirtualBox to identify this machine. Additionally, you can select an ISO image which may be used to install the guest operating system.'

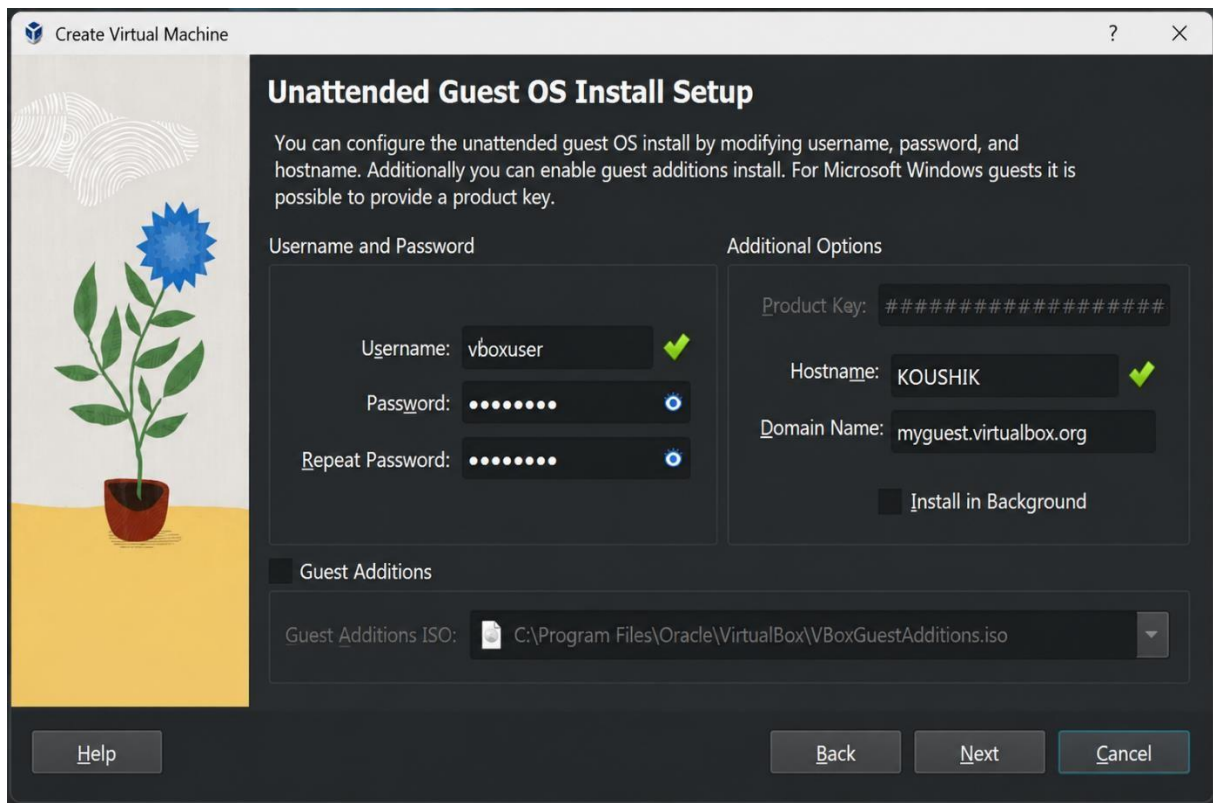
The configuration fields are as follows:

- Name:** koushik (with a green checkmark icon)
- Folder:** C:\Users\marak\VirtualBox VMs
- ISO Image:** <not selected>
- Edition:** (empty dropdown)
- Type:** Microsoft Windows
- Version:** Windows 10 (64-bit) (with a Windows 10 64-bit icon)
- Skip Unattended Installation:** (unchecked checkbox)

An information icon and message state: 'No ISO image is selected, the guest OS will need to be installed manually.'

At the bottom are five buttons: 'Help', 'Expert Mode', 'Back', 'Next' (highlighted), and 'Cancel'.

Step 2 :PROVIDING CREDENTIALS FOR THE VM



Create Virtual Machine

Unattended Guest OS Install Setup

You can configure the unattended guest OS install by modifying username, password, and hostname. Additionally you can enable guest additions install. For Microsoft Windows guests it is possible to provide a product key.

Username and Password

Username: ✓

Password: 🔑

Repeat Password: 🔑

Additional Options

Product Key:

Hostname: ✓

Domain Name:

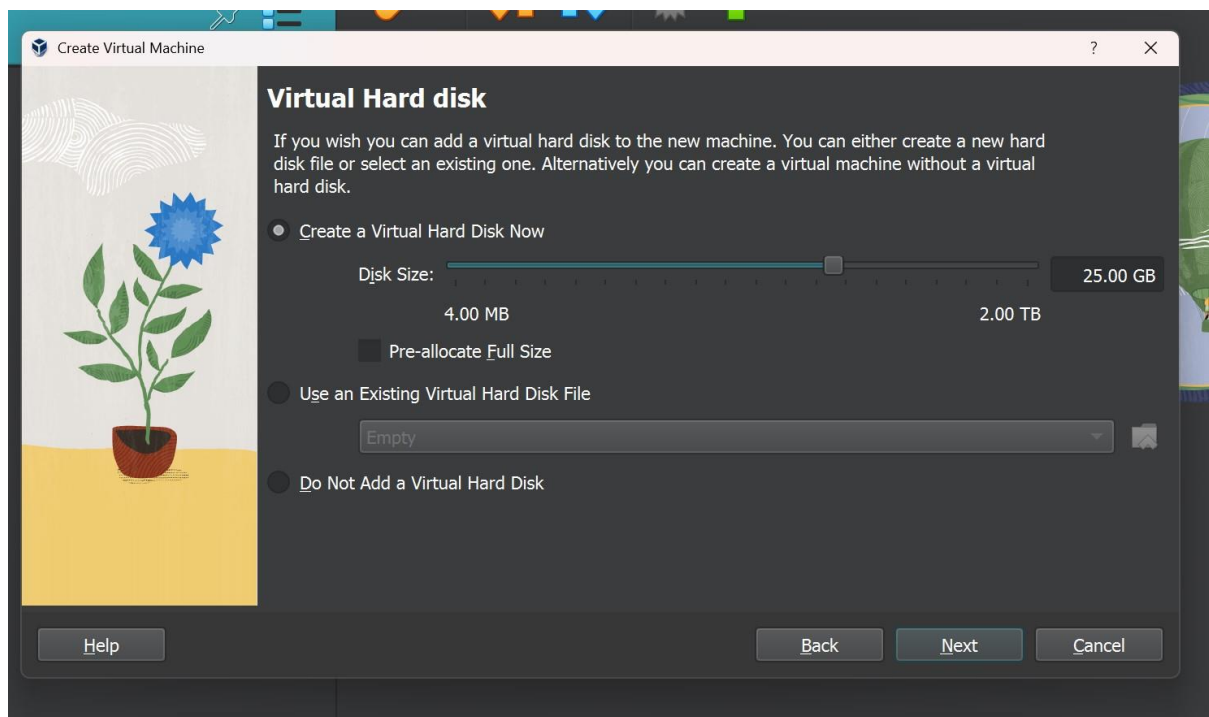
☐ Install in Background

☐ Guest Additions

Guest Additions ISO:

[Help](#) [Back](#) [Next](#) [Cancel](#)

Step 3 : CREATING VIRTUAL HARD DISK WITH REQUIRED CONFIGURATION



Create Virtual Machine

Virtual Hard disk

If you wish you can add a virtual hard disk to the new machine. You can either create a new hard disk file or select an existing one. Alternatively you can create a virtual machine without a virtual hard disk.

☒ Create a Virtual Hard Disk Now

Disk Size: 25.00 GB

4.00 MB 2.00 TB

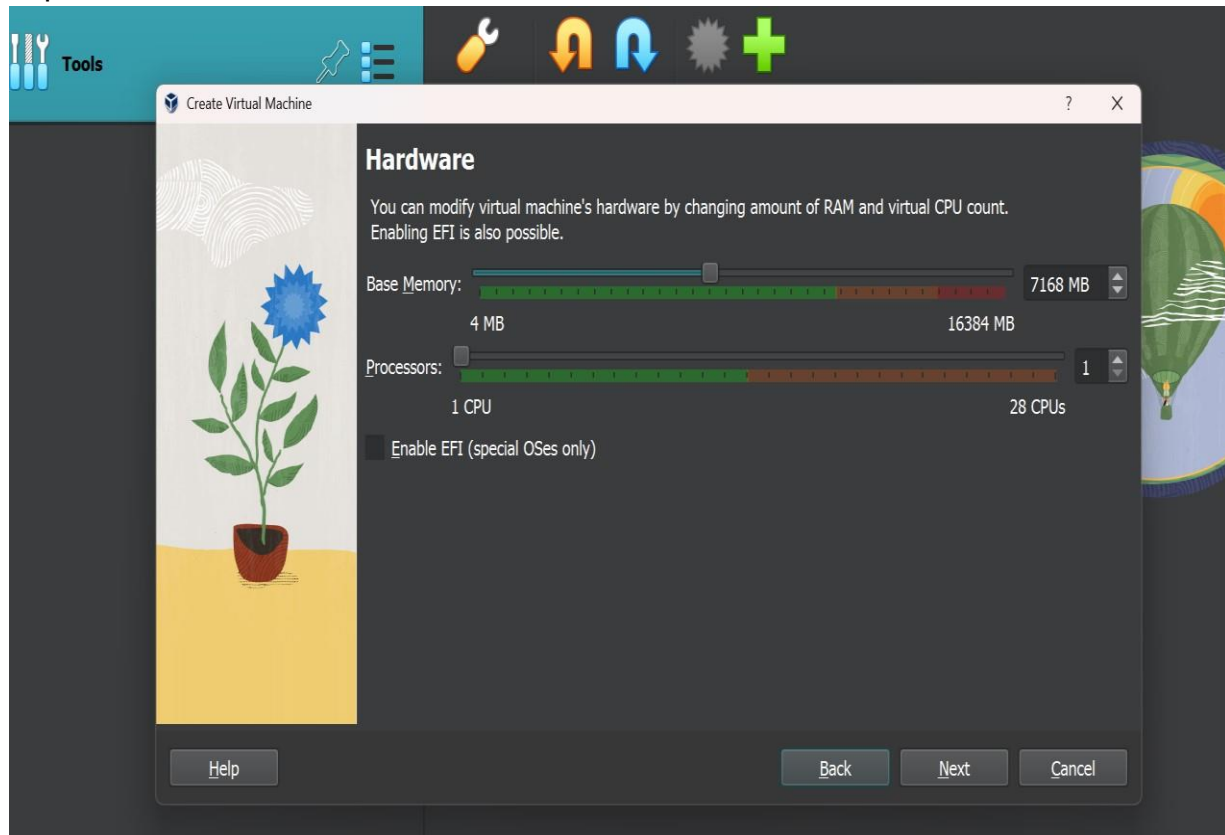
☐ Pre-allocate Full Size

☐ Use an Existing Virtual Hard Disk File

☐ Do Not Add a Virtual Hard Disk

[Help](#) [Back](#) [Next](#) [Cancel](#)

Step 4 : CONFIGURING THE VM

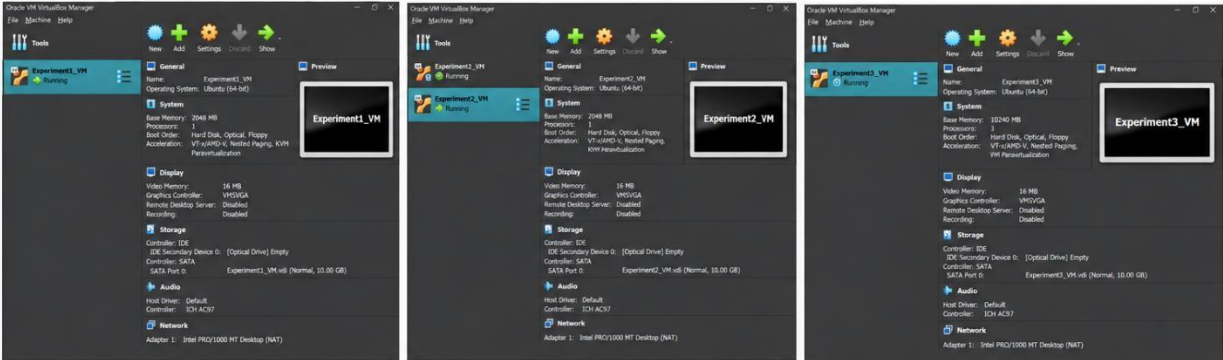


Step 5:- when take different configurations look like these

Record the number of VMs running simultaneously and their configurations.

VM Name	Operating System	RAM (MB)	CPU (Cores)	Storage (GB)	Status
Experiment1_VM	Ubuntu (64-bit)	2048	1	10.00	Running
Experiment2_VM	Ubuntu (64-bit)	2048	1	10.00	Running
Experiment3_VM	Ubuntu (64-bit)	10240	3	10.00	Running

VirtualBox Manager – Running Virtual Machines



Experiment1_VM (2 GB RAM, 1 CPU, 10 GB Storage)

Experiment2_VM (2 GB RAM, 1 CPU, 10 GB Storage)

Experiment3_VM (10 GB RAM, 3 CPU, 10 GB Storage)

Comparison and Maximum Number of VMs

Configuration	Total RAM Used (GB)	Total CPU Cores Used	Total Storage Used (GB)	No. of VMs Running	System Performance	Remarks
VM 1	2 GB	1	10 GB	1	Excellent	Smooth Performance
VM 1 + VM 2	4 GB	2	20 GB	2	Good	Slight Delay Observed
VM 1 + VM 2 + VM 3	14 GB	5	30 GB	3	Acceptable	High RAM Usage
Maximum VMs Running Simultaneously				3 VMs	14 GB RAM, 5 CPU Cores, 30 GB Storage	Beyond this system slows down

Conclusion: In this system configuration, a maximum of 3 virtual machines can be utilized simultaneously with acceptable performance.